# MSG No. 37.185 $C_acc2$ [ Type IV, orthorhombic ]

Table 1: Wyckoff site: `8a`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,2]` |
| 2 | $[0,\ 0,\ z+\frac{1}{2}]$ | `[3,4]` |
| 3 | $[\frac{1}{2},\ \frac{1}{2},\ z]$ | `[5,6]` |
| 4 | $[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}]$ | `[7,8]` |
| 5 | $[\frac{1}{2},\ 0,\ z]$ | `[9,10]` |
| 6 | $[\frac{1}{2},\ 0,\ z+\frac{1}{2}]$ | `[11,12]` |
| 7 | $[0,\ \frac{1}{2},\ z]$ | `[13,14]` |
| 8 | $[0,\ \frac{1}{2},\ z+\frac{1}{2}]$ | `[15,16]` |

Table 2: Wyckoff site: `8b`, site symmetry: `..2'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ \frac{1}{4},\ z]$ | `[1,14]` |
| 2 | $[0,\ \frac{3}{4},\ z]$ | `[2,13]` |
| 3 | $[0,\ \frac{1}{4},\ z+\frac{1}{2}]$ | `[3,16]` |
| 4 | $[0,\ \frac{3}{4},\ z+\frac{1}{2}]$ | `[4,15]` |
| 5 | $[\frac{1}{2},\ \frac{3}{4},\ z]$ | `[5,10]` |
| 6 | $[\frac{1}{2},\ \frac{1}{4},\ z]$ | `[6,9]` |
| 7 | $[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}]$ | `[7,12]` |
| 8 | $[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{2}]$ | `[8,11]` |

Table 3: Wyckoff site: `8c`, site symmetry: `..2'`

| No. | position | mapping |
|---|---|---|
| 1 | $[\frac{1}{4},\ 0,\ z]$ | `[1,10]` |
| 2 | $[\frac{3}{4},\ 0,\ z]$ | `[2,9]` |
| 3 | $[\frac{3}{4},\ 0,\ z+\frac{1}{2}]$ | `[3,12]` |
| 4 | $[\frac{1}{4},\ 0,\ z+\frac{1}{2}]$ | `[4,11]` |
| 5 | $[\frac{3}{4},\ \frac{1}{2},\ z]$ | `[5,14]` |
| 6 | $[\frac{1}{4},\ \frac{1}{2},\ z]$ | `[6,13]` |
| 7 | $[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}]$ | `[7,16]` |
| 8 | $[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}]$ | `[8,15]` |

Table 4: Wyckoff site: `8d`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $[\frac{1}{4},\ \frac{1}{4},\ z]$ | `[1,6]` |
| 2 | $[\frac{3}{4},\ \frac{3}{4},\ z]$ | `[2,5]` |
| 3 | $[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}]$ | `[3,8]` |
| 4 | $[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}]$ | `[4,7]` |
| 5 | $[\frac{3}{4},\ \frac{1}{4},\ z]$ | `[9,14]` |

*continued ...*

Table 4

| No. | position | mapping |
|---|---|---|
| 6 | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | [10,13] |
| 7 | $\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$ | [11,16] |
| 8 | $\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$ | [12,15] |

Table 5: Wyckoff site: 16e, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, z]$ | [1] |
| 2 | $[-x, -y, z]$ | [2] |
| 3 | $\left[-x, y, z+\frac{1}{2}\right]$ | [3] |
| 4 | $\left[x, -y, z+\frac{1}{2}\right]$ | [4] |
| 5 | $\left[x+\frac{1}{2}, y+\frac{1}{2}, z\right]$ | [5] |
| 6 | $\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$ | [6] |
| 7 | $\left[\frac{1}{2}-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$ | [7] |
| 8 | $\left[x+\frac{1}{2}, \frac{1}{2}-y, z+\frac{1}{2}\right]$ | [8] |
| 9 | $\left[x+\frac{1}{2}, y, z\right]$ | [9] |
| 10 | $\left[\frac{1}{2}-x, -y, z\right]$ | [10] |
| 11 | $\left[\frac{1}{2}-x, y, z+\frac{1}{2}\right]$ | [11] |
| 12 | $\left[x+\frac{1}{2}, -y, z+\frac{1}{2}\right]$ | [12] |
| 13 | $\left[x, y+\frac{1}{2}, z\right]$ | [13] |
| 14 | $\left[-x, \frac{1}{2}-y, z\right]$ | [14] |
| 15 | $\left[-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$ | [15] |
| 16 | $\left[x, \frac{1}{2}-y, z+\frac{1}{2}\right]$ | [16] |